

Figure 17.8 Structure of Anther and Pollen grain

Pollen grain: Pollen grains are usually spherical in shape. It has two layered wall. The hard-outer layer is known as **exine**. It has prominent apertures called germpore. The inner thin layer is known as **intine**. It is a thin and continuous layer made up of cellulose and pectin. Mature pollen grains contain two cells, the **vegetative** and the **generative cell**. Vegetative cell contains a large nucleus. The generative cell divides mitotically to form two male gametes.

Gynoecium: Gynoecium is the female part of the flower and is made up of carpels. It has three parts:

1. Ovary
2. Style
3. Stigma

The ovary contains the ovules.

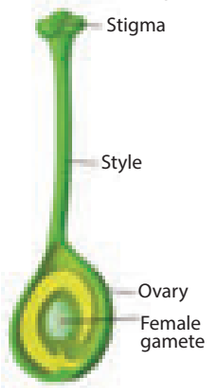


Figure 17.9 Gynoecium

17.3.2 Structure of the Ovule

The main part of the ovule is the **nucellus** which is enclosed by two integuments leaving an opening called as **micropyle**. The ovule is attached to the ovary wall by a stalk known as **funiculus**. **Chalaza** is the basal part.

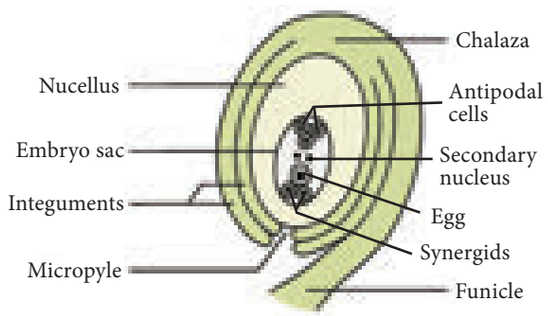


Figure 17.10 Structure of an Ovule

The embryo sac contains seven cells and the eighth nuclei located within the **nucellus**. Three cells at the **micropylar** end form the egg apparatus and the three cells at the **chalaza** end are the antipodal cells. The remaining two nuclei are called **polar nuclei** found in the centre. In the egg apparatus one is the egg cell (female gamete) and the remaining two cells are the **synergids**.

Process of sexual reproduction in flowering plants. It involves:

1. Pollination
2. Fertilization

17.4 Pollination

The transfer of pollen grains from anther to stigma of a flower is called as pollination

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Importance of Pollination

1. It results in fertilization which leads to the formation of fruits and seed.
2. New varieties of plants are formed through new combination of genes in case of cross pollination.

17.4.1 Types of Pollination

1. Self-pollination
2. Cross pollination

Self-pollination (Autogamy)

Self-pollination is also known as autogamy. The transfer of pollen grains from the anther to the stigma of same flower or another flower borne on the same plant is known as self-pollination. e.g. *Hibiscus*.

Advantages of self-pollination

1. Self-pollination is possible in bisexual flowers.
2. Flowers do not depend on agents for pollination.
3. There is no wastage of pollen grains.

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